1) A camera has a lens with a focal length of 5 cm. How far from the film must the lens be for a flower 0.7 m from the lens to be in focus? Describe the image.

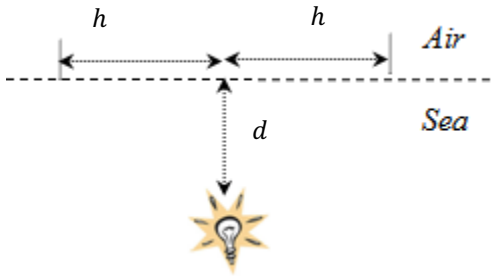
2) A 480-nm light goes through a 2-μm aperture. Find the first three angles of the dark spots.

3) Two monochromatic UV planewave light sources ($\lambda = 420\ nm$), are projected on a screen with 30° between them. Calculate the distance between any two bright spots on the screen.

4) Two monochromatic blue in-phase spherical light sources ($\lambda = 460\ nm$), 100 μm apart are projected on a screen 5 meters away. Calculate the location of the first three bright spots on the screen.

5) At visible light wavelengths, plastic has refractive index of 1.55. Calculate the percentage of power transmitted from air into plastic at normal incidence.

6) A light source that shines uniformly in all directions is submerged at a depth of d=2 meters below the surface of seawater (n=1.35). Assume that the sea surface is flat. Calculate the maximum value of h beyond which an observer at the sea-air interface will not see the light source.



7) Light that emerges out of two polarizers has 65% of the incident light intensity. Is the incident light linearly polarized, circularly polarized or unpolarized? Explain.

8) A laser beam is incident perpendicular to the surface of one of the short faces of a 45–45–90 glass prism. If the refractive index of glass is 1.5, show that the light undergoes total internal reflection when it strikes the long face of the prism. Explain all steps with a sketch and label it.

9) It is required to use the interference phenomenon to determine the density of pollutants in air. Design an experiment for this purpose. Produce a diagram with all labels and descriptions. Explain the theoretical analysis.

10) Draw (to scale with all labels) the interference pattern (intensity versus phase) of 6 (six) plane-wave sources with equal magnitudes ($\boldsymbol{I_o}$) and equal phase differences. Give and explain the interference equation. Show all calculations.

11) A light beam, with polarization direction at 30° from vertical and intensity of 800 W/m^2, passes through a sequence of three polarizers. Measured from the vertical, the transmission axes of the polarizers are at angles of 0°, 30°, and 75°, respectively. What is the intensity of the light when it emerges from each polarizer?

12) With a certain metal plate, shining a beam of red light on the metal causes electrons to be emitted. (a) If we replace the red light by blue light, do we know that electrons will be emitted? (b) If the two beams have the same intensity and are incident on equal areas of the plate, do we

get the same number of electrons emitted per second in the two cases? (c) The plate material is most probably Cs (W=1.85 eV), Na (W=2.4 eV) or Ca (W=2.9 eV). Choose one and explain.

13) Two greenish birds 10 cm apart at distance 300 meters from an observer. What would be a suitable lens for his binoculars for a perfect view? Explain and discuss your result.

14) Find the transmitted power ratio between the red sun light striking sea water normally, and that of violet sun light. Which will penetrate deeper into water? Explain. For water: $n(red) = 1.3314 - j3.48 \times 10^{-8}, n(violate) = 1.34451 - j2.11 \times 10^{-10}$.

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